

## WYNDHAM, Australia

WMO: 942140

Lat: 15.483S

Lon: 128.117E

Elev: 52

StdP: 14.67

Time Zone: 8.00 (E08)

Period: 82-99

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 7             | 59.6       | 61.9 | 20.4                          | 15.4 | 80.6 | 25.1 | 19.2 | 77.8 | 26.8                  | 74.5 | 21.0 | 75.5 | 0.5                   | N/A  | 0.362 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |       |      |       |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|-------|------|-------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%    |      | 2%    |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB    | MCWB | DB    | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)   | (f)  | (g)   | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 11            | 19.0                   | 105.6           | 74.9 | 103.8 | 75.1 | 102.1 | 75.0 | 82.1                | 93.5 | 81.3 | 92.1 | 80.6 | 91.1 | 3.8                  | N/A  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 79.8                            | 154.7 | 86.4 | 78.9 | 150.1 | 85.9 | 78.1 | 146.3 | 85.7 | 46.1          | 93.9 | 45.1 | 92.0 | 44.3 | 91.0 | 94.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |     |  |  |
|-------------------|------|------|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|-----|--|--|
|                   |      |      | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |     |  |  |
| 1%                | 2.5% | 5%   | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max |  |  |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p) |  |  |
| (4) 20.3          | 16.1 | 12.1 | N/A                        | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |
| (5)               |      |      | DB                         | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |
|                   |      |      | WB                         | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |

## Monthly Climatic Design Conditions

|                                                                               |                                                                               | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |
|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |                                                                               | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg                                                                         | 85.0   | 88.3  | 87.0  | 87.6  | 85.4  | 81.7  | 76.4  | 75.6  | 79.5  | 86.2  | 90.7  | 92.0  | 90.3  |      |
|                                                                               | DBStd                                                                         | 6.54   | 4.33  | 3.85  | 3.45  | 3.56  | 3.46  | 4.62  | 4.06  | 4.33  | 2.95  | 2.51  | 3.19  | 4.71  |      |
|                                                                               | HDD50                                                                         | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |      |
|                                                                               | HDD65                                                                         | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |      |
|                                                                               | CDD50                                                                         | 12790  | 1188  | 1037  | 1167  | 1062  | 982   | 793   | 792   | 914   | 1085  | 1262  | 1259  | 1249  |      |
|                                                                               | CDD65                                                                         | 7315   | 723   | 617   | 702   | 612   | 517   | 343   | 327   | 449   | 635   | 797   | 809   | 784   |      |
|                                                                               | CDH74                                                                         | 96846  | 10051 | 8171  | 9678  | 8095  | 6025  | 3400  | 3189  | 4989  | 8361  | 11490 | 12042 | 11355 |      |
|                                                                               | CDH80                                                                         | 54604  | 5762  | 4340  | 5457  | 4437  | 2937  | 1374  | 1253  | 2372  | 4731  | 7142  | 7774  | 7025  |      |
| Wind                                                                          | WSAvg                                                                         | 3.1    | 3.0   | 2.4   | 2.3   | 2.7   | 3.2   | 3.3   | 3.2   | 3.4   | 3.6   | 4.0   | 3.5   | 2.9   |      |
| Precipitation                                                                 | PrecAvg                                                                       | 29.9   | 6.7   | 7.6   | 5.7   | 1.1   | 0.3   | 0.1   | 0.2   | 0.0   | 0.1   | 0.8   | 2.2   | 5.4   |      |
|                                                                               | PrecMax                                                                       | 56.3   | 14.3  | 14.8  | 16.9  | 5.9   | 3.9   | 0.9   | 3.3   | 0.0   | 1.9   | 4.2   | 6.9   | 12.3  |      |
|                                                                               | PrecMin                                                                       | 18.2   | 1.4   | 1.7   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.1   | 0.6   | 0.6   |      |
|                                                                               | PrecStd                                                                       | 8.5    | 3.3   | 3.3   | 4.2   | 1.4   | 0.7   | 0.2   | 0.6   | 0.0   | 0.4   | 1.1   | 1.8   | 3.5   |      |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%                                                                          | DB     | 106.2 | 104.3 | 103.2 | 100.7 | 97.4  | 94.1  | 94.9  | 96.8  | 102.5 | 105.8 | 108.6 | 107.4 |      |
|                                                                               |                                                                               | MCWB   | 73.8  | 78.1  | 76.4  | 72.3  | 70.8  | 71.7  | 67.8  | 68.3  | 71.2  | 73.2  | 73.1  | 75.5  |      |
|                                                                               | 2%                                                                            | DB     | 103.0 | 100.8 | 100.7 | 98.6  | 95.4  | 92.2  | 91.8  | 94.6  | 100.0 | 103.5 | 105.8 | 104.7 |      |
|                                                                               |                                                                               | MCWB   | 76.6  | 78.3  | 75.5  | 72.0  | 70.3  | 69.4  | 66.5  | 68.8  | 70.3  | 72.7  | 74.9  | 75.8  |      |
|                                                                               | 5%                                                                            | DB     | 100.1 | 98.2  | 99.3  | 96.7  | 93.8  | 90.3  | 89.6  | 92.9  | 98.1  | 101.6 | 103.8 | 102.5 |      |
|                                                                               |                                                                               | MCWB   | 77.6  | 78.7  | 75.4  | 71.9  | 70.1  | 68.2  | 65.2  | 67.8  | 70.3  | 72.3  | 75.1  | 76.4  |      |
|                                                                               | 10%                                                                           | DB     | 97.1  | 95.5  | 97.2  | 94.9  | 92.0  | 88.2  | 87.2  | 91.1  | 96.2  | 99.8  | 101.6 | 99.9  |      |
|                                                                               |                                                                               | MCWB   | 78.3  | 78.7  | 75.4  | 71.1  | 69.2  | 66.4  | 63.2  | 66.8  | 70.0  | 72.8  | 75.6  | 77.0  |      |
|                                                                               | Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%   | WB    | 82.8  | 83.1  | 82.5  | 81.9  | 78.7  | 75.5  | 73.2  | 74.0  | 77.3  | 82.4  | 81.9  | 83.3 |
|                                                                               |                                                                               |        | MCDB  | 95.3  | 93.4  | 94.1  | 89.7  | 87.3  | 89.7  | 87.5  | 88.4  | 89.1  | 95.6  | 93.8  | 97.4 |
| 2%                                                                            |                                                                               | WB     | 81.5  | 81.9  | 81.0  | 79.2  | 75.4  | 72.6  | 70.9  | 72.5  | 75.5  | 79.6  | 80.7  | 81.1  |      |
|                                                                               |                                                                               | MCDB   | 93.0  | 92.1  | 90.4  | 87.6  | 87.8  | 86.4  | 84.8  | 88.2  | 88.8  | 92.4  | 93.7  | 92.8  |      |
| 5%                                                                            |                                                                               | WB     | 80.7  | 81.0  | 80.1  | 77.9  | 74.3  | 70.8  | 69.0  | 71.3  | 74.5  | 78.1  | 79.7  | 80.3  |      |
|                                                                               |                                                                               | MCDB   | 91.6  | 90.8  | 89.3  | 87.2  | 87.4  | 85.2  | 83.8  | 87.1  | 89.0  | 91.1  | 92.8  | 91.5  |      |
| 10%                                                                           |                                                                               | WB     | 80.1  | 80.3  | 79.4  | 77.0  | 72.9  | 69.2  | 66.9  | 70.1  | 73.6  | 77.0  | 78.8  | 79.6  |      |
|                                                                               |                                                                               | MCDB   | 90.6  | 89.7  | 89.0  | 86.7  | 86.8  | 84.6  | 82.6  | 85.4  | 89.1  | 90.0  | 91.9  | 91.0  |      |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB                                                                         | MDBR   | 15.9  | 15.3  | 17.2  | 19.1  | 20.3  | 21.3  | 23.1  | 22.9  | 22.1  | 19.9  | 19.0  | 16.9  |      |
|                                                                               |                                                                               | MCDBR  | 20.2  | 19.6  | 20.2  | 20.8  | 21.6  | 22.2  | 24.6  | 24.0  | 23.1  | 22.1  | 21.6  | 20.9  |      |
|                                                                               |                                                                               | MCWBR  | 6.0   | 5.1   | 5.4   | 6.4   | 7.2   | 7.2   | 8.5   | 7.8   | 6.9   | 6.4   | 5.4   | 6.3   |      |
|                                                                               | 5% WB                                                                         | MCDBR  | 16.6  | 16.2  | 16.7  | 15.7  | 18.1  | 20.2  | 22.7  | 21.6  | 20.4  | 19.2  | 18.3  | 16.9  |      |
|                                                                               |                                                                               | MCWBR  | 5.5   | 5.5   | 5.6   | 6.0   | 6.7   | 7.7   | 8.8   | 7.7   | 6.7   | 7.1   | 6.0   | 5.8   |      |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub                                                                          | 0.417  | 0.413 | 0.399 | 0.380 | 0.374 | 0.344 | 0.348 | 0.364 | 0.424 | 0.446 | 0.455 | 0.425 |       |      |
|                                                                               | taud                                                                          | 2.468  | 2.479 | 2.517 | 2.506 | 2.455 | 2.498 | 2.465 | 2.399 | 2.283 | 2.259 | 2.286 | 2.425 |       |      |
|                                                                               | Ebn at Noon                                                                   | 295    | 293   | 291   | 285   | 276   | 280   | 281   | 286   | 279   | 280   | 282   | 292   |       |      |
|                                                                               | Edh at Noon                                                                   | 38     | 37    | 35    | 33    | 33    | 30    | 32    | 36    | 43    | 46    | 45    | 39    |       |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg                                                                        | 1963   | 1959  | 1911  | 1888  | 1711  | 1618  | 1715  | 1953  | 2154  | 2241  | 2207  | 2033  |       |      |
|                                                                               | RadStd                                                                        | 157    | 246   | 222   | 129   | 112   | 71    | 59    | 56    | 85    | 114   | 96    | 127   |       |      |

## Historical Trends

|                             | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (4 neighbors) | N/A   | N/A     | N/A    | +1.17   | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page